PKM lead 1 and m = 4

Lead 1 (bounded D?) on the symmetric PKM tower, and the first m = 4 cells

2026-10-26 · follow-up to `../pkm_symmetric_tower_20261026/` (PR #2632) · instrument `examples/pkm_symmetric_tower.rs` · records `rows.jsonl` (m = 3, sweep t ≤ 7), `rows_n18.jsonl` (m = 3, N = 18 rerun), `rows_m4.jsonl` (m = 4, contended), `rows_m4_uncontended.jsonl` (m = 4, timed alone), `run.log` · graph `d_vs_n.svg` (`plot.py`, recorded values only) · PDF `report.pdf`

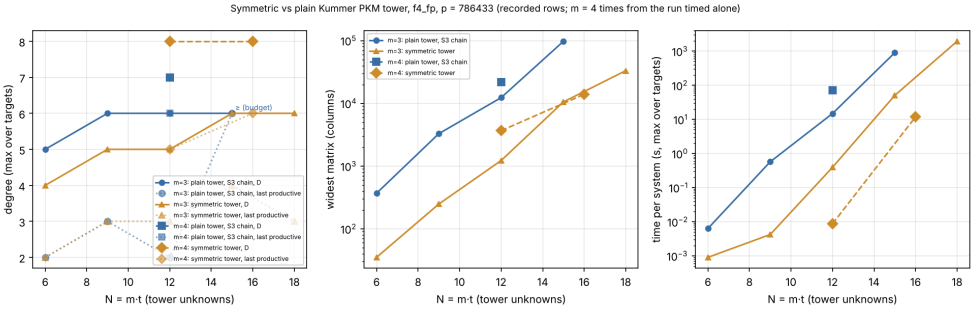
Class: stage diagnostic of the solver axis (AGENTS.md §2). No pipeline is priced, no S, no leaderboard row. Same prime `p = 786433`, curve, `g`, ζ, engine (`f4_fp`, grevlex) and construction as PR #2632.

Verdict

m = 3: the symmetric tower's D is **6 at N = 18**, the same as at N = 15. The sequence over N = 6, 9, 12, 15, 18 is **4, 5, 5, 6, 6**. That is still exactly a staircase that steps up every second cell, so the next step would come at N = 21. The N = 18 cell cannot tell the two readings apart: "D bounded at 6" and "D rises by one every 6 in N" both predict 6 there. **Lead 1 stays open.** The discriminating cell is N = 21, where the two readings predict 6 and 7. It needs more than this container's 13 GB.

m = 4 (new): S_5 written in the e_k is now in the instrument and passes its planted check. At N = 12 the symmetric system runs in 8–9 ms where the plain chain takes 66–70 s (both timed alone on the host), with 5.9× narrower matrices and a last productive degree one lower (5 against 6). Its D is 8 at N = 12 and 16. That is just the degree of $S_5(e)$, its input, so D cannot rank the two arms at m = 4.

Cost still grows with N in both families. From N = 15 to 18 (m = 3) the widest matrix grows 3.1× and the time 39×. Nothing here prices a pipeline.



D and last productive degree, widest matrix and time per system against N; recorded rows only

m = 3: does D stay at 6?

N	symmetric D	last productive	widest matrix	time per system
6	4	2	35	< 1 ms
9	5	3	248	4 ms
12	5	3	1,224	0.4 s
15	6	4	10,532	50 s
18	**6**	3	33,153	1,941 s (one target; alone except one plot render of a few seconds)

Rows N = 6–15 reproduce PR #2632 exactly: same D, widths and field-operation counts.

m = 4: S_5 in the elementary symmetric functions

$S_5(x_1,\ldots,x_4,$ x_R$) =$ Res_u$(S_4(x_1,x_2,x_3,u),\ S_3(u,x_4,x_R))$, where $S_4 =$ Res_v$(S_3(x_1,x_2,v),\ S_3(v,x_3,u))$. It has degree ≤ 8 in each variable. With x_R fixed, the instrument evaluates it on a 9^4 grid (6×6 Sylvester determinants mod p) and recovers it exactly by tensor Lagrange interpolation. It then rewrites it in $e_1..e_4$ with the leading-term algorithm. **Check:** every cell first evaluates every equation at a planted decomposition. The planted decomposition uses four distinct summands from V and x_R computed by chained point additions. The check passes at N = 12 and N = 16.

N	arm	D	last productive	widest matrix	field ops	time per system, alone (contended)
12	plain chain	7	6	21,619	$1.10\cdot10^{11}$	66.4 / 70.2 s (157–159 s)
12	**symmetric**	8	**5**	**3,663**	**$4.01\cdot10^{5}$**	**8.7 / 7.9 ms** (13 ms)
16	**symmetric**	8	**6**	**13,926**	$1.08\cdot10^{10}$	11.8 / 11.7 s (18 s)
20	symmetric	—	—	—	—	aborted under its 6 GB cap

Times are per target (target 0 / target 1). The timed-alone rows are the 20:01Z rerun (rows_m4_uncontended.jsonl), run with nothing else on the host. The contended times in parentheses come from the first run, alongside the m = 3 N = 18 cell, and are kept for the record only (AGENTS.md: wall time is evidence only from uncontended runs).

Both random targets give identical D and widths in every cell, and their field-operation counts agree to within $4 \cdot 10^{-7}$. Both runs give the same D, widths and field-operation counts.

- **D is not the right comparison at m = 4.** The symmetric system's input $S_5(e)$ already has degree 8, so $D \geq 8$ holds trivially. The plain chain's inputs have degree 4, and its D = 7 matches the recorded tower-engine value at N = 12 (encyclopedia §0.1: 6, 7, 7, ≥ 8 at N = 8, 12, 16, 20).
- **The cost is the comparison.** At N = 12 the symmetric system is 5.9× narrower and about 7,600–8,900× faster per system, timed alone (66–70 s against 8–9 ms; the paired per-target ratios are 7,632× and 8,891×). It uses about $2.7 \cdot 10^5$× fewer field operations ($1.1 \cdot 10^{11}$ against $4.0 \cdot 10^5$), and that count does not depend on load. Its last productive degree is one lower (5 against 6).
- From N = 12 to 16 the symmetric arm's last productive degree rises from 5 to 6, and its cost grows by about $2.7 \cdot 10^4$×.

What did not run, and why

- **m = 3, N = 18, first attempt.** The kernel's OOM killer stopped it (exit 137) after 92 CPU-minutes, at about 8.5 GB or more. The operator had started the m = 4 cells alongside it, and together they exceeded the container's 15 GB. It was rerun alone (`rows_n18.jsonl`) with a 13 GB cap and one random target. Because it starts at `--t-min 6`, that target differs from the one a full sweep would draw.
- **m = 3, N = 21** was not attempted after that. It would need more than 13 GB.
- **m = 4, N = 8** cannot be planted. `V` has 4 points, and 200,000 draws found no ordered 4-subset whose chained sum gives an `x_R`. The instrument skips such a cell instead of measuring it without its check.
- **m = 4, N = 16, plain chain** was not run. At N = 12 it already takes 66–70 s alone, with 21,619 columns.
- **m = 4, N = 20, symmetric:** allocation failure under the 6 GB cap (exit 134).
- The first m = 4 times were measured while the m = 3 N = 18 cell held all four cores. They are superseded by the rerun timed alone and are not pooled with it. The contended plain chain was 2.3× slower than alone, and the contended symmetric arm 1.5×. So the contended ratio (12,000×) overstated the uncontended one.
- The resume checkpoint could not be pushed: the container has no S3 token.

Reading

Constant-factor gains in the decomposition oracle's F4 stage are now measured at both m = 3 and m = 4: about 10× narrower matrices at m = 3 and 5.9× at m = 4. The per-system time gain is 20–50× at m = 3 for N = 12–15 (PR #2632), and 7,600–8,900× at m = 4, N = 12 (timed alone). For m = 4 this is the larger effect. The plain chain needs m − 2 = 2 extra partial-sum unknowns and an S_3 chain, while the symmetric system has none, and it sees each unordered summand set once instead of m! = 24 times.

None of this is an index-calculus improvement until it is priced end to end: relations per second against a matched rho, with the §5a band. The next measurements are:

1. **m = 3, N = 21 symmetric.** It decides lead 1's next step (6 against 7). It needs a host with about 32 GB or more.
2. **m = 4, N = 20 symmetric** against the recorded plain D ≥ 8.
3. **Price the m = 4 symmetric oracle** inside the PKM decomposition pipeline at N = 12–16. Whether a per-system gain of about 8,000× moves the pipeline depends on

the share of time the F4 stage takes there. That share has not been measured.